

DECODE LABS
DATA ANALYTICS INTERNSHIP

WEEK 2 –

PROJECT 2

EXPLORATORY DATA ANALYSIS (EDA)

REPORT

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Tool: Microsoft Excel 2016
Dataset Size: 1,200 Records

Executive Summary

This report presents an Exploratory Data Analysis (EDA) of a cleaned e-commerce transaction dataset. The analysis focuses on descriptive statistics, correlation analysis, outlier detection, and trend identification using pivot tables and visualizations.

Problem Statement

The objective was to identify patterns, trends, relationships, and unusual observations within the transaction dataset and transform raw data into actionable business insights.

Methodology

The analysis was conducted using Microsoft Excel 2016. Techniques included descriptive statistics, IQR-based outlier detection, correlation analysis, pivot tables, charts, and box plot analysis.

Dataset Overview

Attribute	Value
Total Records	1,200
Total Variables	14
Tool Used	Microsoft Excel 2016
Analysis Type	Exploratory Data Analysis
Outlier Detection Method	IQR Method

Descriptive Statistics Summary

Statistic	Quantity	UnitPrice	ItemsInCart	TotalPrice
Count	1200	1200	1200	1200
Mean	2.95	356.41	5.49	1053.97
Median	3	364.21	5	823.62
Minimum	1	11.39	1	11.39
Maximum	5	699.93	10	3456.40
Q1	2	186.06	4	410.52
Q3	4	521.57	7	1578.48
IQR	2	335.51	3	1167.96
Upper Bound	7	1024.83	11.5	3330.41
Outliers	0	0	0	8

Correlation Analysis

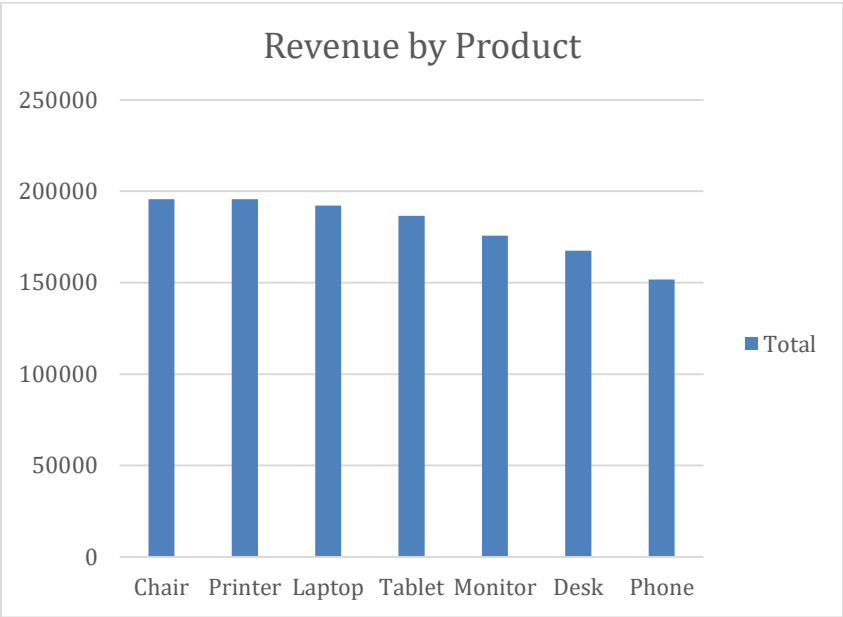
Variables	Correlation	Interpretation
UnitPrice vs TotalPrice	0.717	Strong Positive
Quantity vs TotalPrice	0.615	Moderate Positive
ItemsInCart vs TotalPrice	0.393	Weak Positive

Outlier Analysis

Using the IQR method, 8 high-value outlier transactions were identified. These transactions exceeded the calculated upper bound of 3,330.41.

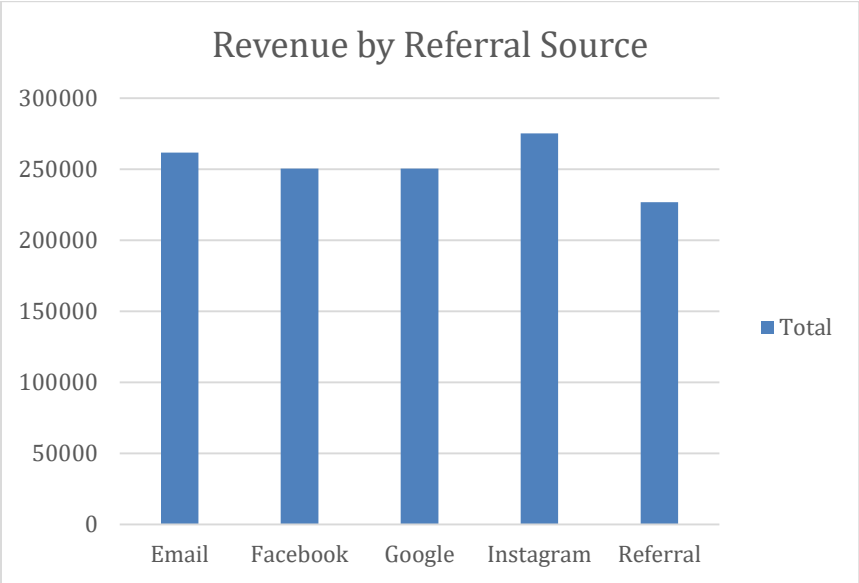
Trend Analysis & Visualizations

Figure 1: Revenue by Product Chart



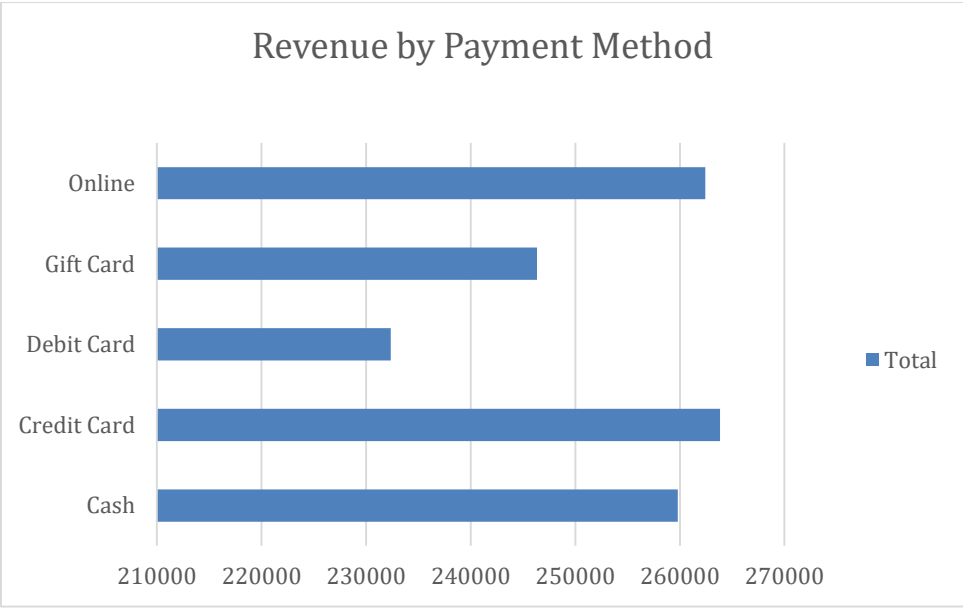
Observation: Chair generated the highest revenue (\$195,620.11), while Phone generated the lowest revenue (\$151,722.39). Revenue distribution was relatively balanced across product categories.

Figure 2: Revenue by Referral Source Chart



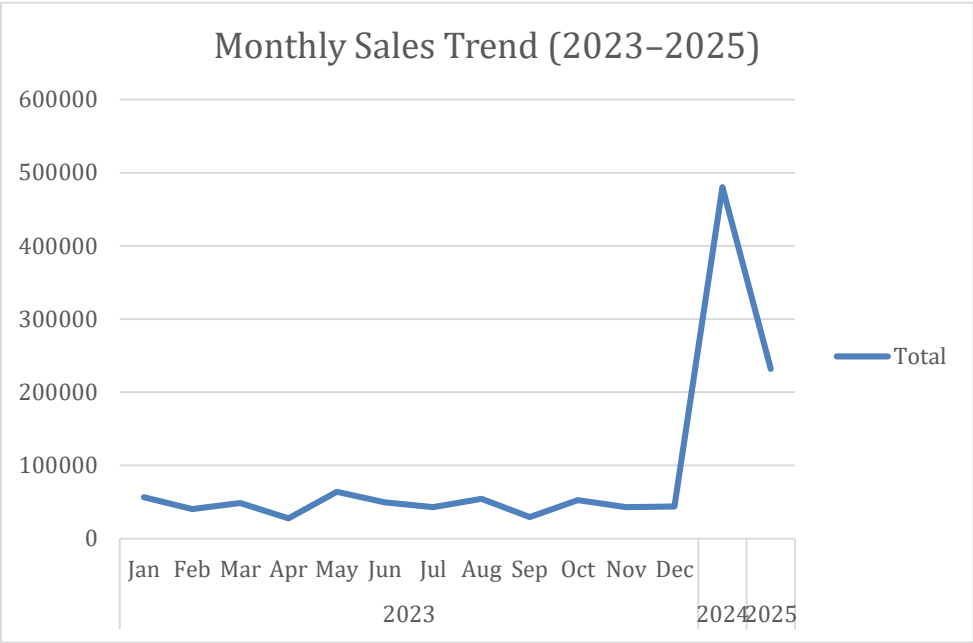
Observation: Instagram generated the highest revenue (\$275,285.45), indicating that it is the most effective customer acquisition channel.

Figure 3: Revenue by Payment Method Chart



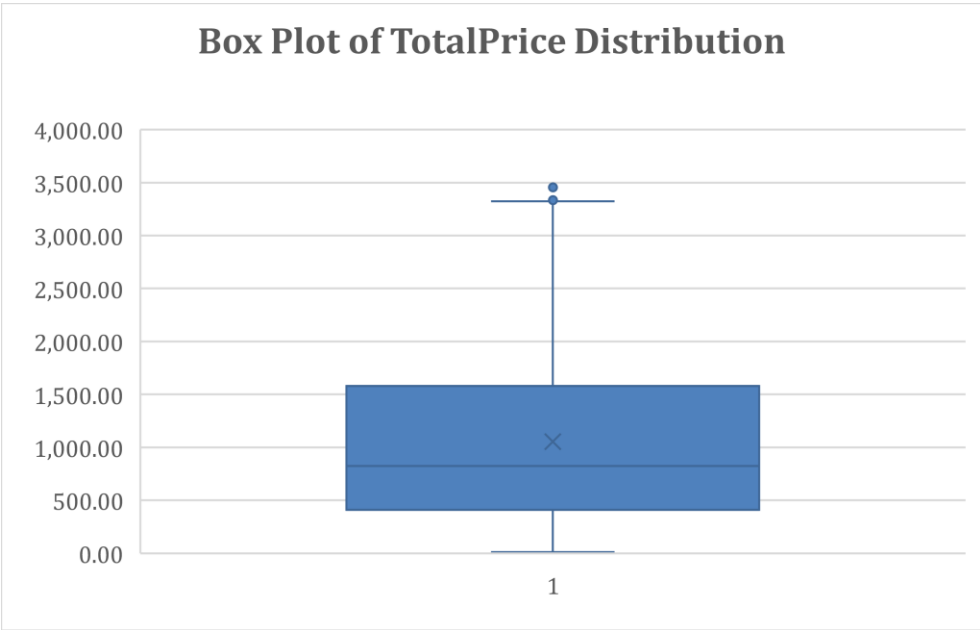
Observation: Credit Card transactions generated the highest revenue, suggesting customers prefer card-based payments for higher-value purchases.

Figure 4: Monthly Sales Trend Chart



Observation: Revenue fluctuated across the observed period, with noticeable increases in later periods, indicating changing customer purchasing behavior.

Figure 5: Box Plot of TotalPrice Distribution



Observation: The box plot confirms the presence of high-value outliers above the upper whisker, supporting the IQR analysis which identified 8 outlier transactions.

Key Findings

- Chair generated highest product revenue.
- Instagram generated highest referral-source revenue.
- Credit Card generated highest payment-method revenue.
- UnitPrice showed strongest correlation with TotalPrice.
- Eight outlier transactions were identified.
- Average transaction value was 1,053.97.
- TotalPrice exhibited a positively skewed distribution.
- No outliers were detected in Quantity, UnitPrice, or ItemsInCart.
- Revenue varied across products and acquisition channels.
- Customer spending patterns showed substantial variation.

Recommendations

- Focus marketing efforts on high-performing acquisition channels such as Instagram.
- Investigate high-value outlier transactions to identify premium customer behavior.
- Leverage pricing strategies as UnitPrice has the strongest impact on TotalPrice.
- Monitor product performance regularly using dashboard-based reporting.

Conclusion

The EDA successfully identified trends, correlations, outliers, and business performance indicators from 1,200 transaction records. The findings provide valuable insights for future decision-making and performance monitoring.